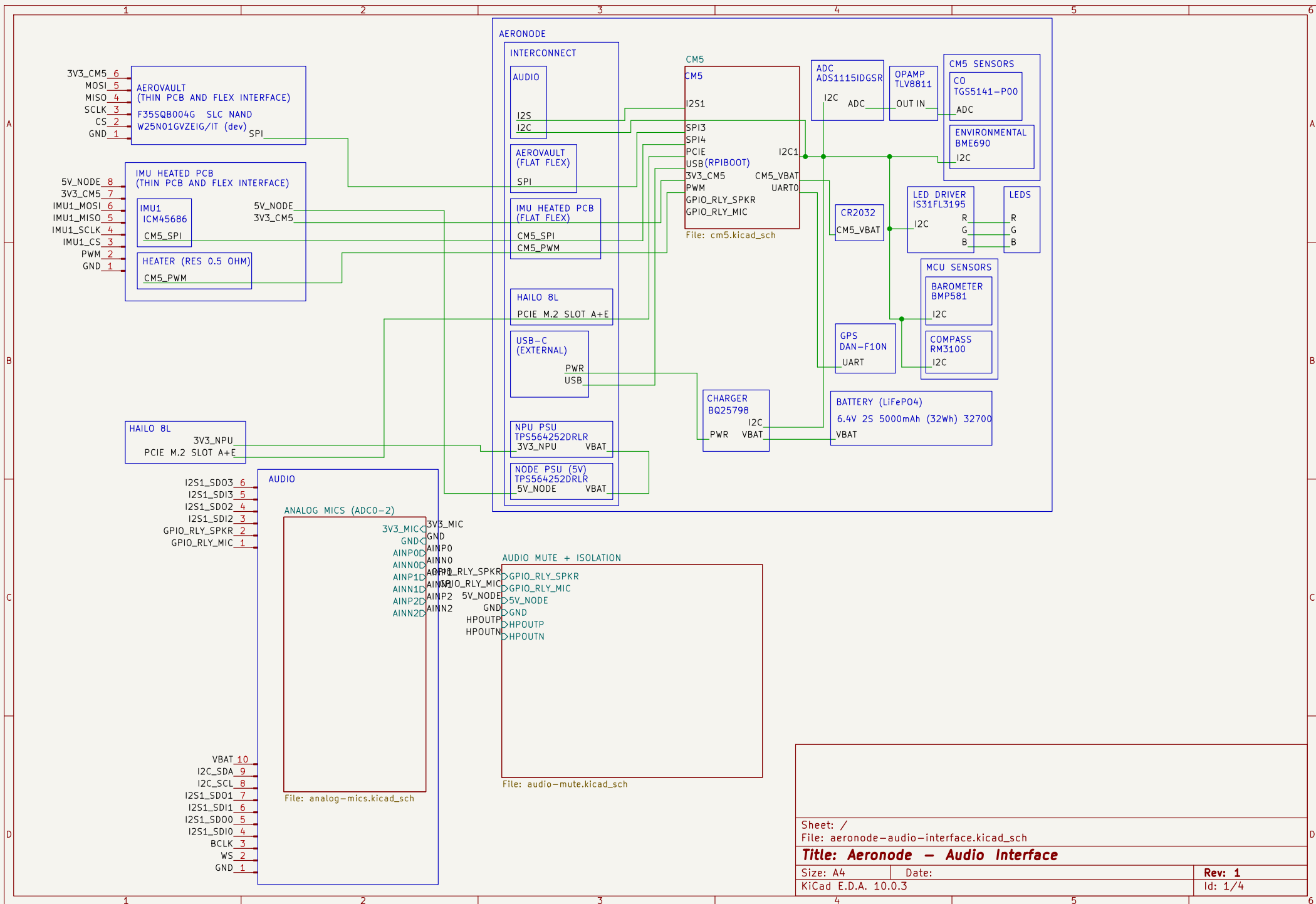
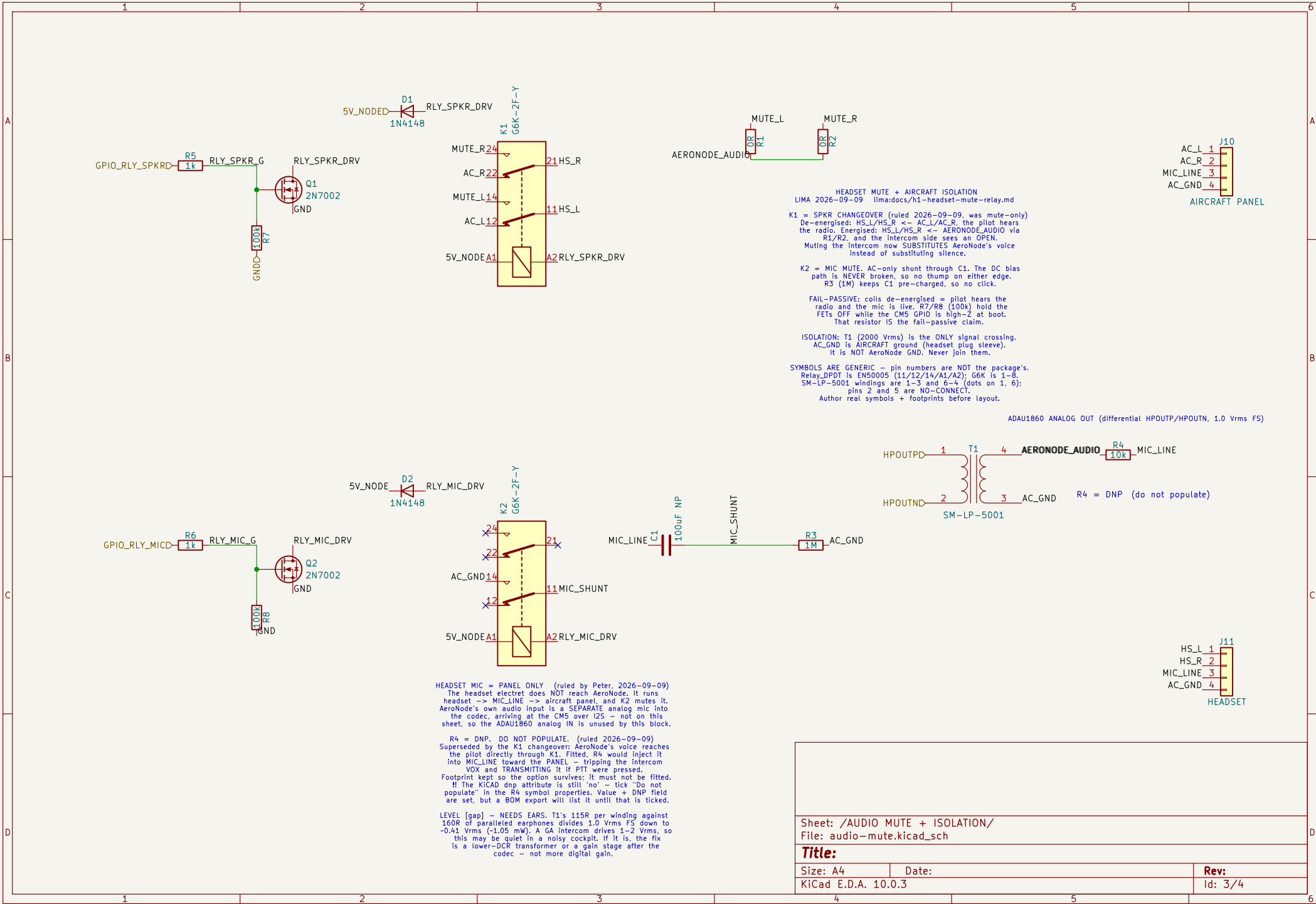




1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

Sheet: /CM5/ File: cm5.kicad_sch		
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ALL THREE MICS ARE ON THE AERONODE SIDE. They are our mics, biased from our rail and referenced to our GND, so they never touch AC_GND. The isolation boundary on audio-mute.kicad_sch is untouched - no transformer is needed here.

FRONT END, per channel: Rn (100R) + Cn (1uF) filter the supply at the connector. C2n / C3n couple OUT and the mic's local GND into AINPn / AINNn - pseudo-differential, so cable ground noise is rejected. This mirrors the EVB, which puts 22uF in series with BOTH legs and no bias network (UG-2017 Figure 8).

USE AMPLIFIED MIC MODULES, NOT BARE CAPSULES.

The ADAU1860 has NO MICBIA5 pin, and its PGA is only 0-24 dB against a 0.49 Vrms single-ended full scale. A bare electret gives a few mV; x16 lands about -24 dBFS at best and the rest has to come from digital gain, which lifts the noise floor with it. Modules must output a few hundred mV and run off 3V3 MIC.

3V3_MIC MUST BE ITS OWN QUIET RAIL – not the CM5's 3V3. ANC noise performance is set by the mic supply. [gap] No LDO is specified yet, which is why 3V3_MIC shows one deliberate "label not connected" ERC error on the parent sheet.

ANC RUNS ONLY WHILE K1 IS ENERGISED (ruled by Peter, 2026-09-09)
K1 stays a CHANGEOVER. AeroNode's DAC reaches the earcup only when AeroNode is the active audio path, and ANC runs then – not continuously. This was raised as a blocker: Peter ruled it is the intended behaviour, so K1 does NOT sum. Summing was investigated and is NOT a free swap – see section 20. In short: OR-ing the residual signal with the DAC output is not a free swap. At the panel level AeroNode retains depends entirely on the panel's headphone output impedance (~28 dB into 10R, -6.5 dB into 600R). Do not "improve" this to a summing node without re-reading that section.
LATENCY: at 48 kHz the ADC→DSP→DAC loop is 1 ms, too slow for feedforward above a few hundred Hz. Raising the RASP→DAC loop rate to 192 kHz characterises it at 768 kHz for exactly this.

T1 is not a bandwidth limit here: its 200 Hz–4 kHz spec is Bourns' 600R telecom condition: into 160R from a low-Z drive the corner computes to 3.4 Hz.

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